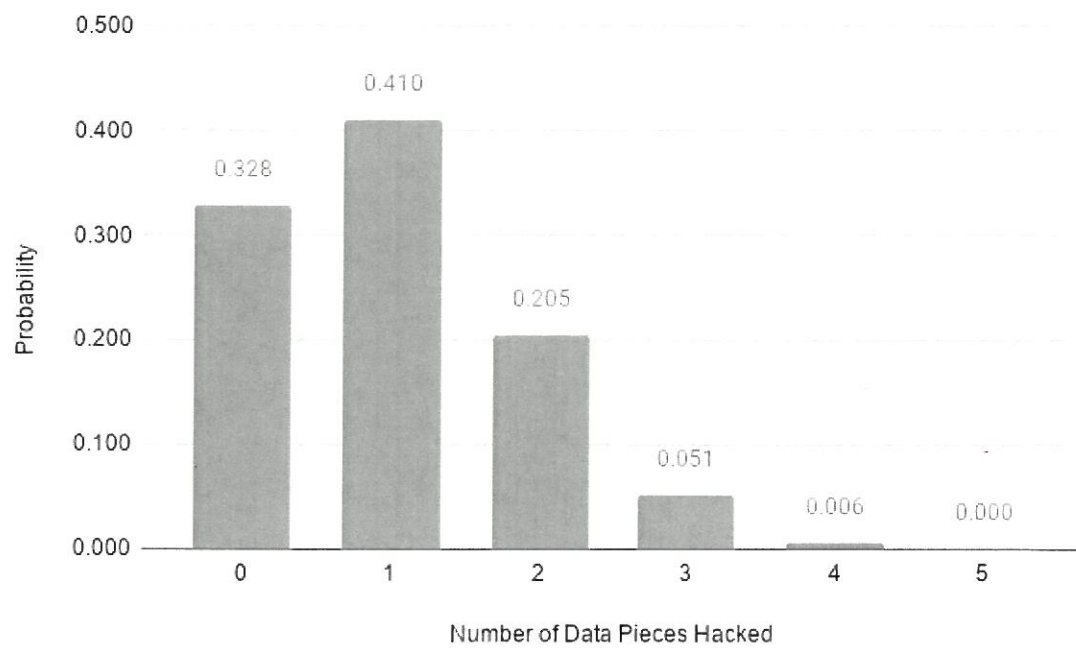


**LECTURE 33: MATH1061/7861**  
**TOPICS: BINOMIAL COEFFICIENTS**  
**DATE: THURSDAY MAY 14, 2026**

Learning objectives:

- (1) Understand the Binomial Theorem.
- (2) Learn how to evaluate coefficients in binomial expansions.



Q1: You are sending 5 pieces of data across a network. Hackers are on standby to hack your data. Despite your best effort, they have a 20% chance of success. What is the probability that:

(a) None of the 5 pieces of data is hacked?

$$\left(\frac{4}{5}\right) \times \left(\frac{4}{5}\right) \times \dots \times \left(\frac{4}{5}\right)$$

$$= \left(\frac{4}{5}\right)^5 \approx 32\%$$

(b) Exactly one is hacked?

$$5 \times (0.2) \times (0.8)^4 \approx 41\%$$

(c) Exactly two are hacked?

$$\binom{5}{2} = 10$$

$$\binom{5}{2} \times \left(\frac{4}{5}\right)^3 \times \left(\frac{1}{5}\right)^2 \approx 20.5\%$$

(d) Exactly three are hacked?

$$\binom{5}{3} \times \left(\frac{1}{5}\right)^3 \times \left(\frac{4}{5}\right)^2 \approx 5.1\%$$

(e) Exactly four are hacked?

$$\binom{5}{4} \times \left(\frac{1}{5}\right)^4 \times \left(\frac{4}{5}\right) \approx 0.6\%$$

(f) All five are hacked?

$$\left(\frac{1}{5}\right)^5 \leftarrow \text{less than } 0.01\%$$

Binomial Theorem:

$$\binom{5}{3} = \binom{5}{2}$$

$$(a+b)^n = \sum_{k=0}^n a^k b^{n-k} \binom{n}{k}$$

Q3: Compute the coefficient of  $x^3$  in the expansion of  $(2x-3)^5$ .

$$(2x)^3 (-3)^2 \binom{5}{3} = 8 \times 9 \times 10 \times x^3$$

So coeff is 720

What is the coefficient of  $x^2$ ?

$$\binom{5}{2} (2x)^2 (-3)^3 = -(10 \times 4 \times 27) x^2$$

$$= -1080 x^2$$

$$\text{coeff} = -1080.$$

$$p \in [0, 1].$$

Q2: Suppose now you are sending  $n$  pieces of data and hackers have a probability  $p$  of successfully hacking each piece. What is the probability of having exactly  $k$  hacked files?

Let's call this  ~~$x_k$~~   $x_k$

$$x_k = \binom{n}{k} p^k (1-p)^{n-k}$$

What can you say about the sum:

$$\cancel{p_0 + p_1 + \dots + p_n} \quad x_0 + x_1 + \dots + x_n = 1$$

$$\text{I.e.} \quad \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} = 1$$

$$\text{Binomial Theorem: } (a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$